

Electricity Resource Adequacy by Exact Probabilistic Convolution

How much interconnection does Belgium need to keep the lights on? — a compound-distribution model on real grid data, validated against Monte Carlo

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Abstract

Will there be enough electricity when demand is highest? This note answers that question for Belgium and, crucially, checks its own answer. Each of about a hundred power stations is either running or unexpectedly broken down; combining all of them gives the probability distribution of how much power is available at any moment, which is compared against a full year of real demand to obtain two standard reliability figures — **LOLE** (the expected hours per year of shortage) and **EENS** (the expected energy not served). The available-power distribution is built *exactly* by an FFT convolution and then independently reproduced by a Monte-Carlo simulation; the two agree to 0.2%, which is what makes the answer trustworthy. On its own power stations alone, Belgium would face 22.3 hours of shortage per year against a target of 3 — but only about 1 GW of reliable imports closes that gap, and the first gigawatt is worth far more than the next. A sensitivity check shows the risk is driven far more by how high demand climbs than by how often plants break down. The contribution is not a new adequacy model or reliability metric — LOLE, EENS, forced-outage rates and probabilistic aggregation are all well established. It is a *computational validation framework*: an exact FFT convolution and an independently written Monte-Carlo simulation are required to certify one another before any number is accepted — the answer is *certified*, *not asserted*, and then used to say what it would take to fix.

1. The question, in plain terms

Electricity is unusual: it cannot be stored in large amounts, so at every instant the power flowing onto the grid must match the power being drawn off it. If demand ever exceeds what the available power stations can supply, something has to give — in the worst case, the lights go out somewhere. *Resource adequacy* is the study of how often that can happen, and whether a country has built enough generation to keep it rare.

The industry measures this risk with a single number, **LOLE** — *loss-of-load expectation* — the expected number of hours per year in which supply cannot meet demand. Countries plan to a standard: Belgium's is 3 hours per year. The difficulty is that these shortage hours are *rare* and *random* — they depend on cold, still, dark winter evenings coinciding with several power stations happening to be offline at once — so you cannot simply read the risk off historical data. You have to build a probabilistic model.

Belgium is a good test case: a mid-sized country, heavily connected to its neighbours, that has publicly debated whether it has enough of its own generation. The question this note poses is deliberately sharp — *how much reliable import capacity does Belgium's own fleet need to meet the 3-hour standard?* — and it falls out of an exact model. The guiding principle, shared with the companion studies on pension longevity and clinical-trial drug supply, is that a number is only worth reporting if two independent methods agree on it.

This note therefore makes no claim to a new adequacy method. Its subject is the *discipline* around the number: build the model exactly one way, reproduce it a completely independent way, and accept a result only when the two agree — an agreement that, as §3 recounts, had to be *earned* by uncovering a hidden modelling assumption rather than simply assumed. The engineering conclusions that follow (§4) are what that verified model then lets one say.

2. The model — how it works

2.1 Adding up unreliable power stations

Start with the supply side. For adequacy, what matters about each power station is whether it *could* deliver power if called on — so the relevant states are **available** (able to run, whether or not it happens to be generating at that moment) or on **forced outage** (unexpectedly broken, delivering nothing). A plant sitting idle because demand is low still counts as available: adequacy asks about *capability*, not about who is dispatched. Engineers summarise

the risk of an unplanned breakdown with the *forced-outage rate* — the fraction of time a plant is unexpectedly out. A gas plant might be out 6% of the time; a nuclear unit, 10%.

This leaves out one real state: **planned maintenance**, when a plant is deliberately taken offline for scheduled work. The saving grace is that operators schedule maintenance into *low-demand* seasons — spring, summer, quiet weekends — precisely to keep the fleet available through the winter-evening peaks where adequacy risk lives. So at the hours that actually drive the risk, the two-state picture is a fair approximation; we return to this assumption among the model’s limits (§6).

The total power available to the country is the sum, over all stations, of “rated power if running, zero if out.” Because each station is a small independent random contribution and we want the distribution of their sum, this is a **compound distribution** — mathematically the same object as an insurer’s total claims (many independent risks added up), here an aggregate *capacity* rather than an aggregate loss. Its exact distribution is obtained by *convolving* the individual plants together, which a Fast Fourier Transform (FFT) performs in a single efficient step:

$$\mathbf{f}_A = \mathcal{F}^{-1} \left[\prod_i \mathcal{F}[\mathbf{f}_{A_i}] \right],$$

where A_i is plant i ’s contribution. No approximation, no simulation — the exact probability of every possible total-available-power level. Convolution by FFT is chosen over the classical recursive (Panjer-style) approach for speed: it costs on the order of $n \log n$ operations against the recursion’s n^2 , which matters when the power axis is finely gridded.

For Belgium in 2023 the fleet is 102 dispatchable units totalling 13,118 MW. Convolving them gives an available-power distribution centred on 12,154 MW — the *expected* amount running, once typical breakdowns are accounted for.

Fuel	Units	Capacity (MW)	Typically out
Nuclear	7	4,937	10%
Gas	49	5,512	6%
Hydro	15	1,395	3%
Oil / liquid	11	182	12%
Other	20	1,094	7%

2.2 Demand — taken straight from reality

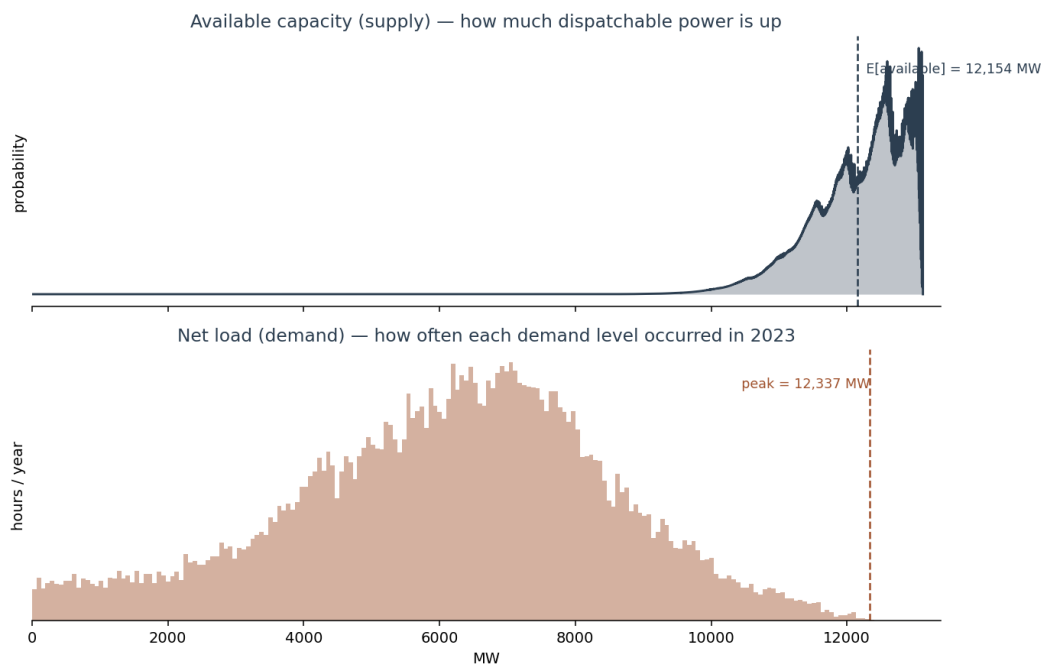
Now the demand side. What the power stations actually have to cover is not total electricity demand but **net load** — total demand *minus* whatever wind and solar happen to be producing, since renewable output isn’t dispatchable and simply reduces what the conventional fleet must supply. We take this net load directly from Elia’s real 2023 record — every quarter-hour of the year, 35,037 values — rather than fitting it to an idealised curve. Using the real data keeps the model honest and sidesteps any argument about which demand distribution to assume.

The figure below shows the two sides together. They are drawn separately and deliberately *not* overlaid: supply (top) is tightly bunched near 12 GW — most plants run most of the time — while demand (bottom) is spread across 4–9 GW with a thin tail reaching the annual peak of 12,337 MW.

The key observation is already visible. The peak of demand (12,337 MW) sits *just above* the 12,154 MW *expected* (mean) available supply. These two are not strictly comparable — one is an annual extreme, the other an average — so this margin is only a first indicator, not the risk itself. The actual risk arises when a high-demand hour coincides with a *low* capacity draw (several plants out at once, in the left tail of the supply distribution), and it is precisely this coincidence that the full LOLE calculation below integrates over. But the closeness of peak demand to mean supply already signals that, on its own fleet, Belgium runs a thin margin.

2.3 Turning the two sides into a risk number

For each of the 35,037 quarter-hours we ask: what is the probability that available power falls below that hour’s demand? The available-power distribution answers this directly. Adding those probabilities across the year



Two SEPARATE distributions on the same MW axis — deliberately not overlaid, because whether supply and demand are low/high in the SAME hour cannot be read from marginals alone. TOP: supply is tightly concentrated near 12 GW (most units available most of the time). BOTTOM: demand is spread 4–9 GW, with a thin tail reaching ~12.3 GW. They occupy almost non-overlapping ranges — which is why loss-of-load is rare. Exactly WHEN supply falls below demand is what the next figure (load-duration curve) shows directly.

Figure 1: Supply and demand as two separate distributions on the same power axis. A shortage can occur only in the thin region where the high tail of demand meets the low tail of supply.

(in hours) gives LOLE; adding the expected size of each shortfall gives EENS. On the domestic fleet, with no imports:

$$\text{LOLE} = 22.3 \text{ hours/year}, \quad \text{EENS} \approx 11,000 \text{ MWh/year}.$$

Against the 3-hour standard this looks alarming — but it is a *domestic-only* figure (no imports), and the gap to the standard is exactly what Belgium’s interconnection is worth (§5). It is also worth seeing *when* the risk occurs, because that is not obvious:

The risk is extraordinarily concentrated. Of 35,037 quarter-hours, the great majority carry essentially no risk at all; the entire 22.3 hours piles into a few freezing, wind-still, dark evenings in January, February, November and December. In spring and summer, net load frequently goes *negative* — wind and solar alone exceed total demand — and adequacy risk is simply zero. This concentration is characteristic of adequacy — it is a rare-tail problem, which is exactly why average-based intuition fails and a full probabilistic model is needed.

3. Validation — why you can trust the number

Here is the part that distinguishes this work. A model that produces a plausible-looking LOLE is not the same as a *correct* one, and the rare-tail nature of the problem makes errors easy to hide. So the exact FFT result is checked against a completely independent method: a **Monte-Carlo simulation** that randomly switches each plant on or off many thousands of times and simply counts the shortage hours. If the exact calculation is right, the simulation must converge onto it.

The two agree to 0.2%. But that agreement was *earned*, and how it was earned is itself a result. The first time the two methods were compared they disagreed by 15–20% — the exact method said 22.3 hours, the simulation said 18.9. A fixed disagreement that does not shrink with more simulation is not random error; it means the two methods are quietly computing *different things*. Tracking it down revealed the cause: the exact calculation treated each quarter-hour’s plant availability as independent, whereas the first simulation had frozen each plant’s state for a whole year at a time. These are two different physical assumptions about how long a breakdown lasts. Reconciling the simulation to match the exact model brought them into agreement.

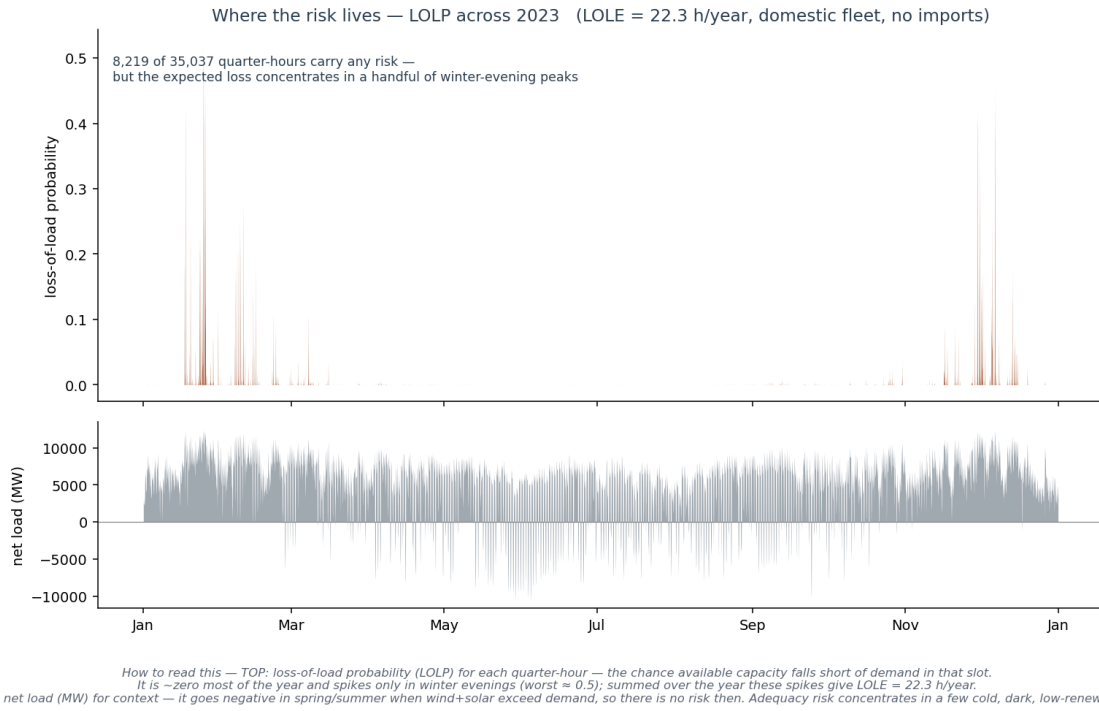
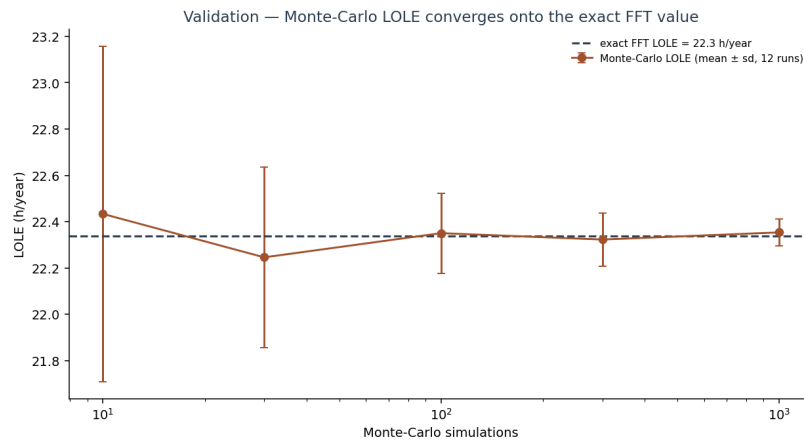


Figure 2: Where the risk lives across 2023. Almost every hour is risk-free; the danger concentrates in a handful of cold winter evenings (top). The bottom panel shows why — net load peaks in winter and goes negative in summer, when wind and solar exceed demand.



The adequacy engine computes LOLE two independent ways. The exact value (dashed) comes from FFT convolution of the 102 units into the available-capacity distribution. Monte Carlo (points) samples unit up/down states directly; as simulations grow its LOLE converges onto the exact value, error bars shrinking. Agreement across independent methods is the certification.

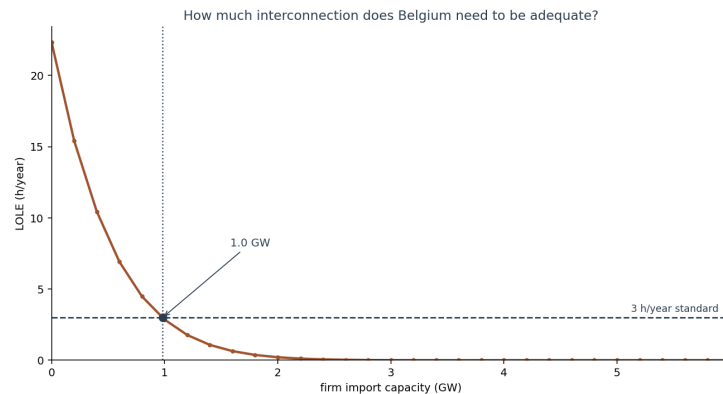
Figure 3: Validation: as the Monte-Carlo simulation runs more trials, its LOLE estimate converges tightly onto the exact FFT value, error bars shrinking. Agreement between two independent methods is the certification.

That episode is more than a fixed bug. The two assumptions — outages lasting an instant versus a whole year — bracket reality, since a real breakdown lasts *hours to days*. Outages that persist across a single tight evening make consecutive risky hours correlated, which *worsens* the tail and pushes LOLE *up*. So the 22.3-hour figure, built on the independent-hours assumption, is best read as a **lower bound** — a floor, not a ceiling. Independent validation did not merely confirm a number; it exposed a hidden modelling choice and told us which direction the truth lies in. That is the whole point of the “certified, not asserted” discipline.

4. The results

4.1 How much interconnection does Belgium need?

Belgium does not, of course, rely on its own plants alone — it imports power from France, the Netherlands and Germany, especially at peak. We can price that dependence exactly. A firm (always-available) import capacity of I megawatts simply lowers the domestic requirement by I every hour, so its effect on LOLE is read straight off the same model with demand shifted down. Sweeping I upward from zero traces the curve below.



LOLE against firm (perfectly reliable) import capacity. With no imports Belgium's domestic fleet gives 22.3 h/year; adding firm interconnection drives it down. The curve crosses the 3 h/year reliability standard at the marked point — the interconnection Belgium needs to be adequate. This is an OPTIMISTIC bound: real imports aren't firm (neighbours are often tight at the same cold, calm hours), so the true requirement is higher.

Figure 4: LOLE against firm import capacity. With no imports Belgium sits at 22.3 hours/year; roughly 1 GW of firm interconnection brings it down to the 3-hour standard, after which the curve flattens.

The result is striking in two ways. First, the crossing: Belgium's fleet misses the standard by 19 hours, yet only about **1 GW of firm imports** — a modest amount for a system this size — closes the gap entirely. Belgium is not badly under-built; it sits right on the edge, and a small reliable margin fixes it. The reason a single gigawatt does so much is that shortages occur in very few hours and are mostly *shallow* — demand exceeds supply by a relatively small amount — so a modest block of firm capacity removes most shortage events outright. Second, the *shape*: the curve plunges over the first gigawatt and then flattens to almost nothing by 2.5 GW. So the **first** gigawatt of interconnection is worth enormously more than the third — the value of imports is front-loaded, precisely the kind of insight that guides where investment should go.

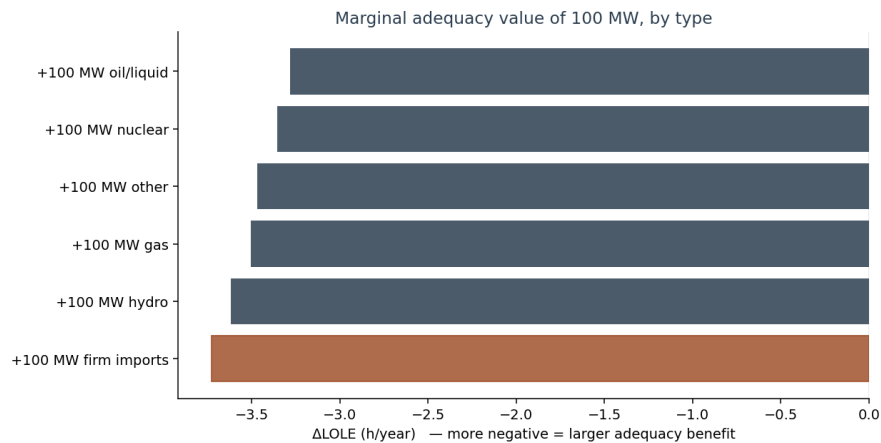
One honest qualification, built into the model: these are *firm* imports, assumed always available. Real interconnection is worth somewhat less, because a cold snap that strains Belgium usually strains its neighbours at the same time — so imports are least available exactly when they are most needed. The 1-GW figure is therefore an optimistic floor; the true requirement is higher, which is why Belgium contracts several gigawatts of interconnection plus a capacity-support mechanism, rather than exactly one. The model gives the floor; prudence adds the margin.

4.2 What actually drives the risk?

Because the aggregation is *exact under the stated model*, we can compare scenarios without simulation noise blurring the differences — a genuine advantage over Monte Carlo when the effect being measured is small. Two questions, two answers.

First: *what is 100 MW of new capacity worth, by type?* Rebuilding the model with one extra 100-MW unit of each kind gives:

The technologies land close together — all buy roughly 3.5 hours/year of reduction — with the ordering set



Change in LOLE from adding 100 MW, each carrying its own technology's forced-outage rate (firm imports = perfectly reliable). Lower-outage, always-available capacity is worth more — firm imports and low-FOR hydro shave the tight winter tail hardest; higher-outage plant less so. Differences are exact (each an FFT re-convolution, no sampling noise).

Figure 5: The adequacy value of an extra 100 MW, by technology. All buy a similar reduction; what matters is reliability — firm imports and low-outage hydro shave the winter peak hardest, high-outage plant slightly less.

purely by reliability: firm imports (never out) at the top, then hydro (3% out), down to oil/liquid (12% out). The lesson: what a megawatt is worth for adequacy depends on how dependably it is *there* on a freezing evening, not on its fuel.

Second, and more revealing: *what moves LOLE most?* Nudging each driver by 1% and measuring the response (its elasticity):

Driver	Sensitivity of LOLE (per 1% change)
Peak / total demand	+21.6
Plant breakdown rates	+2.4
Total capacity	-17.5

Demand is the giant. LOLE is roughly **nine times more sensitive to how high demand climbs than to how often plants break down**. In plain terms: a 1% rise in electricity demand hurts reliability about nine times more than a 1% worsening of plant reliability. The adequacy threat is *demand growth* — think electrification, heat pumps, data centres — far more than aging plant. That single comparison reframes the policy conversation, and it comes for free from the exact model.

5. What the model does not include — four honest limits

Every one of these makes the headline figure *conservative*, and every one is a clear next step rather than a hidden assumption.

No imports in the baseline. The 22.3-hour figure is Belgium *alone*; the interconnection curve above then prices what imports are worth. **A single weather year** (2023). Adequacy risk lives in extreme weather, so a full official study (such as ENTSO-E's or Elia's) replays *many* historical weather years to capture the worst; our figure should be read as "for a 2023-like year." **Textbook breakdown rates.** Each plant's outage rate comes from standard planning tables by fuel type, not from its own observed history. **Forced outages only.** The model counts unplanned breakdowns but assumes no *planned* maintenance at peak — reasonable, since operators schedule maintenance away from winter, but a maintenance-timing model would make it exact. The three natural refinements follow directly: model imports with their own availability (and correlation with Belgian cold snaps), replay many weather years, and estimate breakdown rates from the actual outage record. Stated together, these boundaries make 22.3 hours a precise, well-posed measurement: *how adequate is Belgium's own fleet, in a 2023-like year, at textbook breakdown rates, before it calls on its neighbours?*

6. Reproducibility

Every input traces to public data — demand, wind, solar and the plant list from **Elia Open Data** (no account needed), plant outages from **ENTSO-E** (free registration) — and the model is *built* by exact convolution rather than fitted to anything, so every number can be explained and the whole study reproduces without re-downloading. The complete implementation, all figures, the validation and the results are in the accompanying notebook, `resource_adequacy.ipynb`.